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Course Title : Computer and Digital Forensics

Lab Title: Lab 3 Data Carving with XXD, Binwalk and Scalpel

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Executive Summary

This investigation focused on data carving and file recovery from forensic disk images using industry-standard digital forensic tools. The objective was to demonstrate the recovery and examination of files when filesystem metadata may be deleted, unavailable, or unreliable.

The investigation involved hexadecimal analysis of a JPEG file using xxd, extraction and examination of metadata using ExifTool, filesystem and deleted-file analysis using The Sleuth Kit, embedded-content analysis using Binwalk, and file carving using Scalpel. Deleted files were identified and recovered from a FAT12 forensic image, while a FAT16 USB forensic image was examined at the partition, filesystem, and sector levels.

Scalpel was configured to identify JPEG files using their file signatures and successfully carved 17 JPEG files from the USB forensic image. The carved files were recorded in Scalpel's audit.txt, providing a repeatable record of the recovery process.

Cryptographic hashes, including SHA-256, were generated at appropriate stages to support evidence integrity and verification. The investigation demonstrated that useful file content can remain recoverable from forensic images even when normal filesystem metadata is missing or files have been deleted.

All analysis was performed on working copies or extracted evidence, while the original forensic evidence was preserved to maintain forensic integrity.

Lab Objective

The objectives of this laboratory exercise were to:

1. Understand the principles and importance of data carving and file recovery in digital forensics.
2. Examine binary data using hexadecimal analysis and identify common file signatures and 3. headers/footers.
Analyze JPEG files by identifying SOI (FF D8) and EOI (FF D9) markers.
4. Reconstruct a binary file from hexadecimal data and verify its integrity using cryptographic hashes.
5. Use The Sleuth Kit to examine forensic disk images, filesystem structures, deleted files, inodes, sectors, and recoverable data.
6. Use Binwalk to identify embedded data and file structures within a forensic image.
7. Configure and use Scalpel to carve deleted JPEG files directly from a raw USB forensic image.
8. Verify recovered evidence using MD5 and SHA-256 hashing where appropriate.
9. Document the forensic procedures, findings, screenshots, and limitations in a professional and repeatable manner.

Tools and Resources Used

The following tools and resources were used during the investigation:

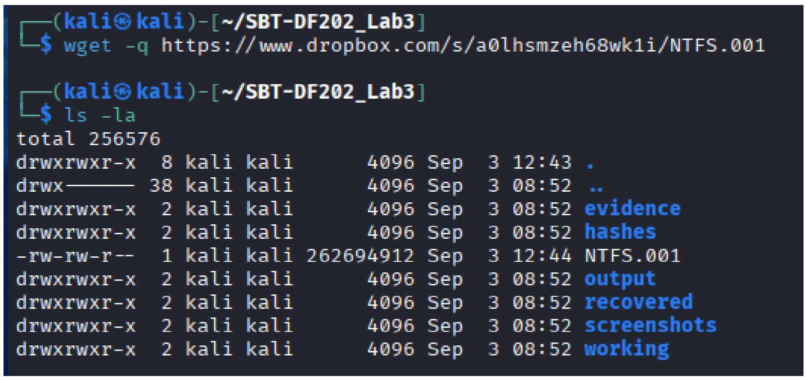
Software and Forensic Tools
Kali Linux — forensic analysis operating environment.
The Sleuth Kit (TSK) — filesystem, partition, inode, sector, and deleted-file analysis.
Autopsy Forensic Browser — forensic case management and evidence examination.
Scalpel 1.60 — signature-based file carving and recovery.
Binwalk — identification of embedded file structures and signatures.
ExifTool — extraction and examination of JPEG metadata.
xxd — hexadecimal examination and binary file reconstruction.
7-Zip (7z) — extraction and examination of the compressed forensic evidence archive.
Linux hashing utilities (md5sum, sha256sum) — cryptographic integrity verification.
file utility — identification and validation of recovered file types.
strings — examination of readable strings within binary and recovered files.
Evidence and Resources
Ch01InChap01.dd — FAT12 forensic disk image used for filesystem and deleted-file analysis.
J_ub_law.jpg — JPEG sample used for hexadecimal, metadata, reconstruction, and Binwalk analysis.
120M.7z — compressed forensic evidence archive containing the USB image.
usb_fat_carving.001 — FAT16 USB forensic image used for Scalpel file carving.
/etc/scalpel/scalpel.conf — Scalpel configuration file containing the enabled JPEG carving signatures.
Evidence Integrity

Cryptographic hashing, particularly SHA-256, was used to document and verify the integrity of forensic evidence and recovered data throughout the investigation.

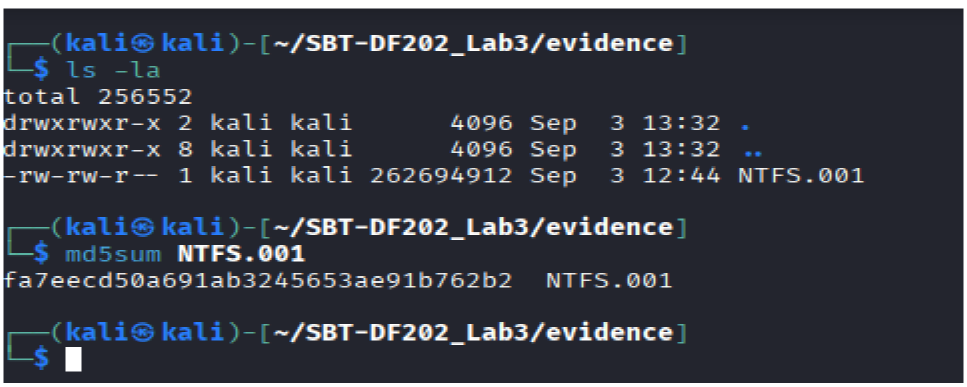
Methodology

The investigation followed a structured forensic workflow. Evidence integrity was established using cryptographic hashing before analysis. The JPEG sample was examined using hexadecimal analysis, metadata extraction, file reconstruction, and Binwalk. The forensic disk images were analyzed using The Sleuth Kit to identify filesystem structures and deleted files. The USB image was examined and processed with Scalpel for signature-based JPEG carving. Recovered files were validated and documented using file identification, hashing, audit logs, and screenshots.

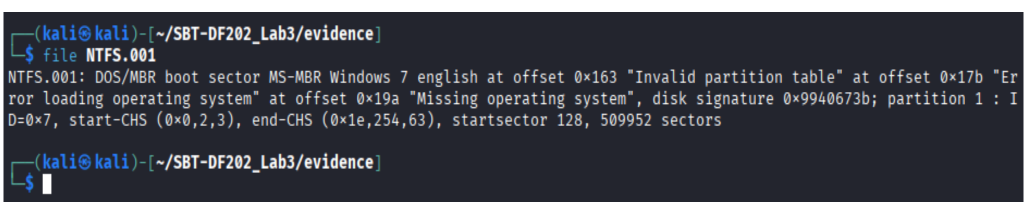
Screenshots and Evidence



```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ wget -q https://www.dropbox.com/s/a0lhsmzeh68wk1i/NTFS.001

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls -la
total 256576
drwxrwxr-x  8 kali kali      4096 Sep  3 12:43 .
drwx------ 38 kali kali      4096 Sep  3 08:52 ..
drwxrwxr-x  2 kali kali      4096 Sep  3 08:52 evidence
drwxrwxr-x  2 kali kali      4096 Sep  3 08:52 hashes
-rw-rw-r--  1 kali kali 262694912 Sep  3 12:44 NTFS.001
drwxrwxr-x  2 kali kali      4096 Sep  3 08:52 output
drwxrwxr-x  2 kali kali      4096 Sep  3 08:52 recovered
drwxrwxr-x  2 kali kali      4096 Sep  3 08:52 screenshots
drwxrwxr-x  2 kali kali      4096 Sep  3 08:52 working
```



```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ ls -la
total 256552
drwxrwxr-x 2 kali kali      4096 Sep  3 13:32 .
drwxrwxr-x 8 kali kali      4096 Sep  3 13:32 ..
-rw-rw-r-- 1 kali kali 262694912 Sep  3 12:44 NTFS.001

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ md5sum NTFS.001
fa7eecd50a691ab3245653ae91b762b2  NTFS.001

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$
```



```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ file NTFS.001
NTFS.001: DOS/MBR boot sector MS-MBR Windows 7 english at offset 0x163 "Invalid partition table" at offset 0x17b "Er
ror loading operating system" at offset 0x19a "Missing operating system", disk signature 0x9940673b; partition 1 : I
D=0x7, start-CHS (0x0,2,3), end-CHS (0x1e,254,63), startsector 128, 509952 sectors

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$
```

Part A

Download an image to kali

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensic
s/file_carving/usb_file/J_ub_law.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls -la
total 1688
drwxrwxr-x  8 kali kali    4096 Sep  3 16:29 .
drwx------ 38 kali kali    4096 Sep  3 08:52 ..
drwxrwxr-x  2 kali kali    4096 Sep  3 13:32 evidence
drwxrwxr-x  2 kali kali    4096 Sep  3 08:52 hashes
-rw-rw-r--  1 kali kali 1692150 Sep  3 16:29 J_ub_law.jpg
drwxrwxr-x  2 kali kali    4096 Sep  3 08:52 output
drwxrwxr-x  2 kali kali    4096 Sep  3 08:52 recovered
drwxrwxr-x  2 kali kali    4096 Sep  3 08:52 screenshots
drwxrwxr-x  2 kali kali    4096 Sep  3 08:52 working

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls J_ub_law.jpg -l
-rw-rw-r-- 1 kali kali 1692150 Sep  3 16:29 J_ub_law.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ display J_ub_law.jpg
```

Record the original hashes

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ md5sum J_ub_law.jpg | tee hashes/J_ub_law_original.md5
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ sha256sum J_ub_law.jpg | tee hashes/J_ub_law_original.sha256
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ cat hashes/J_ub_law_original.md5
cat hashes/J_ub_law_original.sha256
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg
```

Original Evidence Hash Verification

Before analysis, the original J_ub_law.jpg file was verified using MD5 and SHA-256 cryptographic hashing. The recorded hashes establish a baseline for demonstrating the integrity of the evidence throughout the forensic examination.

File: J_ub_law.jpg
Size: 1,692,150 bytes
MD5: 83a360ac7f7e0ca318e5bfe39f95f137
SHA-256: 238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468
The hash values were saved in the hashes/ directory for later comparison with the reconstructed image.

Examine the JPEG header

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ......Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ..............@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ................
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ................
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ................
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ................
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ................
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ................
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  ..............(.
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ..........1...!.

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ xxd J_ub_law.jpg | tail
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae  ..=R..0......"..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7  -.b.z...z.WPk.c.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981  .U.$..d.Mj......
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9  U.....c`A....0:.
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd  ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80  )O.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45  .....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1  .*.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9                           .Q....
```

Note: The xxd utility was used to examine the binary structure of J_ub_law.jpg. The beginning of the file contained the JPEG Start of Image (SOI) signature FF D8 at offset 0x00000000. This was immediately followed by FF E1, indicating an APP1 segment containing Exif metadata.

The end of the file contained the JPEG End of Image (EOI) signature FF D9, observed at the final bytes of the hexadecimal output. The presence of both the SOI and EOI markers confirms the expected JPEG file boundary structure.

These signatures are important in digital forensics because they can be used to identify and carve JPEG data even when normal file-system metadata is unavailable or damaged.

Create a plain hexadecimal dump

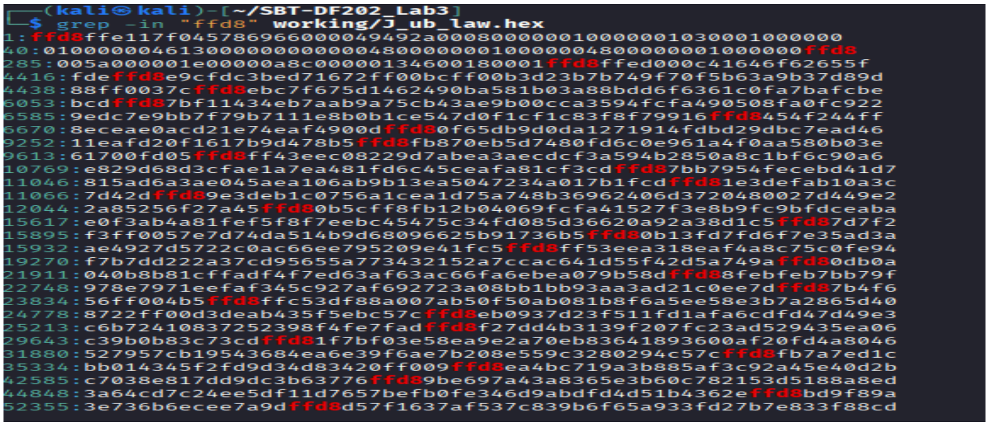
```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ xxd -p J_ub_law.jpg > working/J_ub_law.hex

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ head working/J_ub_law.hex
ffd8ffe117f04578696600004949 2a000800000010000001030001000000
```

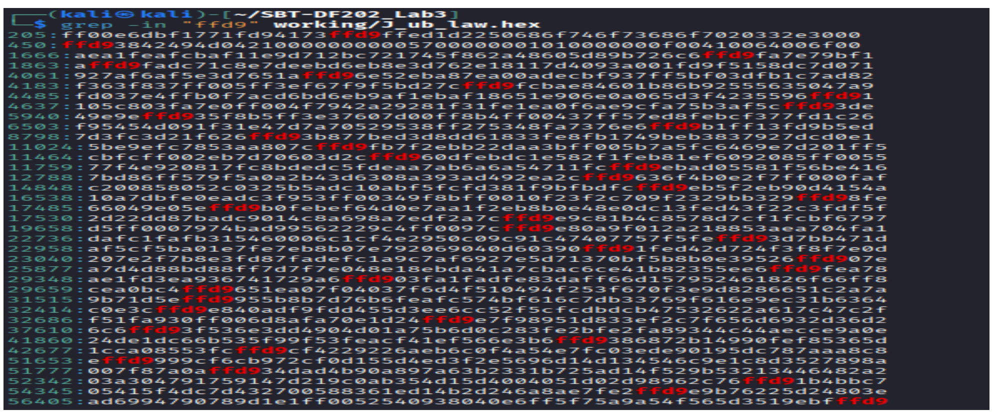
```
4013000001010300010000000d00c0000020103000300000 0ce000000000601
```

A plain hexadecimal representation of J_ub_law.jpg was created using the xxd -p utility and saved as working/J_ub_law.hex. Examination of the dump showed the JPEG SOI signature FFD8 at the beginning of the file and the EOI signature FFD9 at the end.

The plain hexadecimal format removes the address offsets and ASCII display used by the standard xxd output, making it suitable for searching directly for hexadecimal file signatures and for subsequent reconstruction of the binary file.

JPEG Signature Search

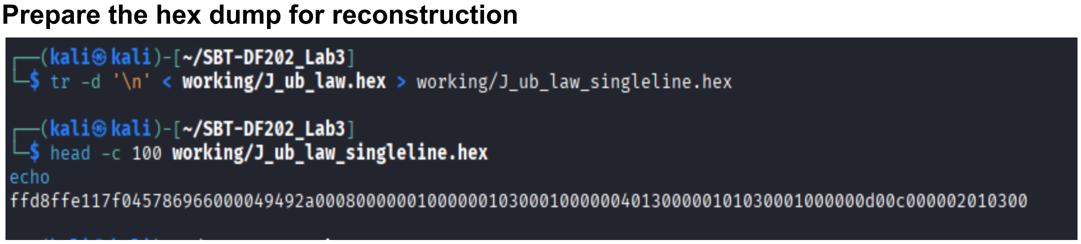




The plain hexadecimal dump was searched for the JPEG SOI (FFD8) and EOI (FFD9) signatures. Multiple occurrences of both byte sequences were identified within the dump. This demonstrates that signature patterns may occur inside compressed JPEG data and therefore cannot automatically be interpreted as file boundaries.

The valid JPEG boundaries were confirmed from the original xxd output: FFD8 occurs at offset 0x00000000 as the Start of Image marker, while FFD9 occurs at the end of the file as the End of Image marker.

Prepare the hex dump for reconstruction



Reconstruct the JPEG

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ xxd -r -p working/J_ub_law_singleline.hex working/J_ub_law_reconstructed.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls -lh working/J_ub_law_reconstructed.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  3 17:25 working/J_ub_law_reconstructed.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ file working/J_ub_law_reconstructed.jpg
working/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16,
width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, ori
entation=upper-left], baseline, precision 8, 1920x1080, components 3
```

Verify the hashes

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ md5sum J_ub_law.jpg working/J_ub_law_reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137  working/J_ub_law_reconstructed.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ sha256sum J_ub_law.jpg working/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  working/J_ub_law_reconstructed.jpg

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
```

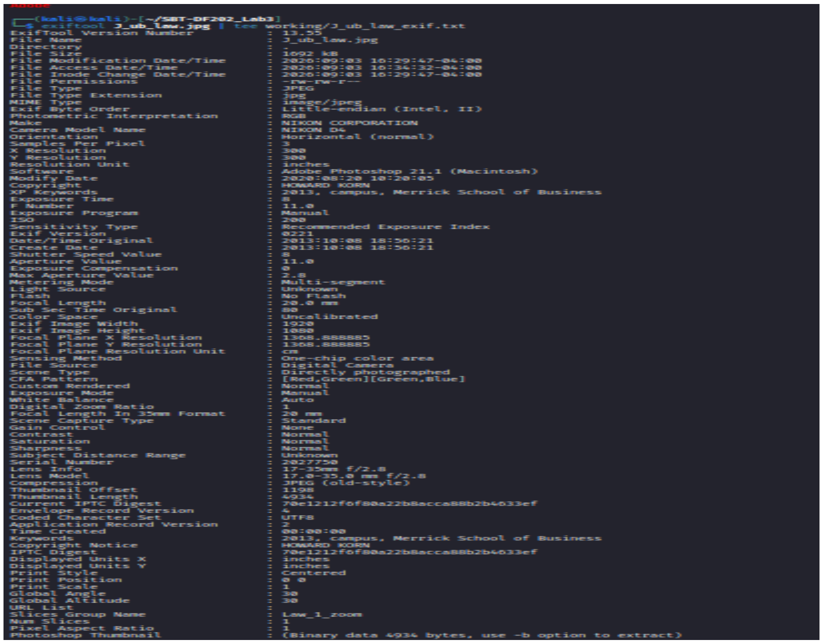
Hexadecimal Reconstruction and Integrity Verification
The JPEG file was converted to a hexadecimal representation and reconstructed using xxd -r -p. The reconstructed file was identified as valid JPEG image data with Exif metadata, including Nikon D4 camera information.
Integrity verification produced identical hashes for both files:

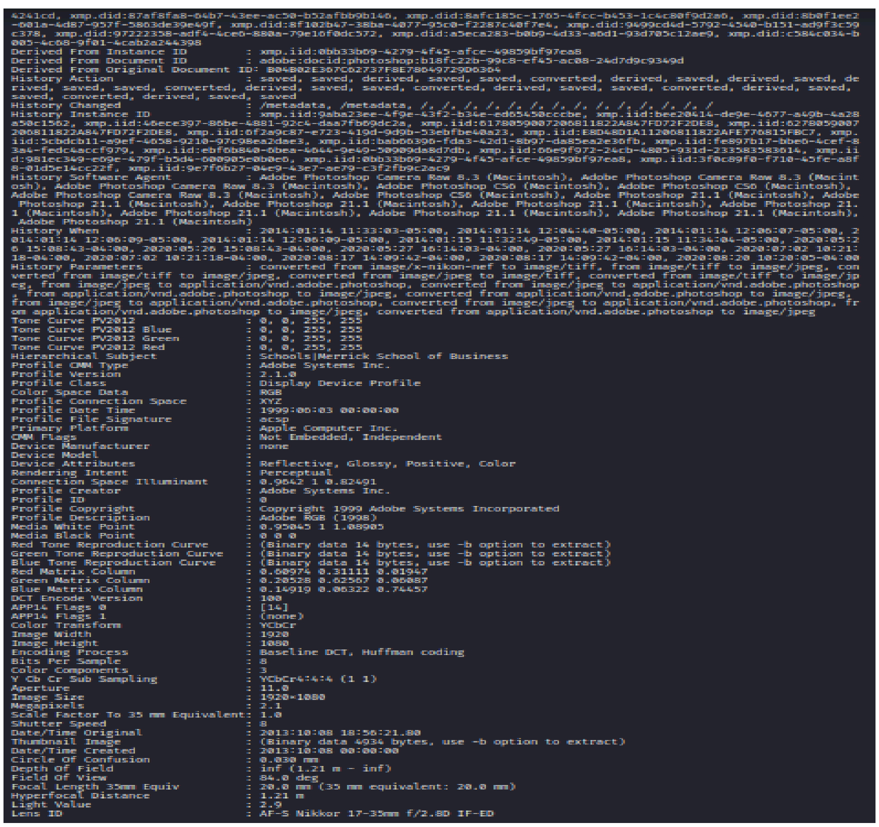
MD5: 83a360ac7f7e0ca318e5bfe39f95f137
SHA-256: 238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468

The matching MD5 and SHA-256 values confirm that the reconstructed JPEG is bit-for-bit identical to the original evidence file.

Part B

Strings and Metadata Examination





ExifTool was used to examine embedded JPEG metadata. The file was identified as a JPEG produced by a NIKON D4 camera. The metadata records an original capture date of 8 October 2013 at 18:56:21.80, with an exposure of 8 seconds, f/11 aperture, ISO 200, and a 20 mm focal length.

The metadata identifies HOWARD KORN as the copyright holder and contains the keywords "2013, campus, Merrick School of Business." The image was processed using Adobe Photoshop 21.1 (Macintosh), with metadata showing subsequent editing and conversion activity through 2020.

The metadata also contains a camera serial number, lens information, raw file name, document identifiers, and editing history. These fields may provide useful investigative leads but should be interpreted cautiously because metadata can be modified during image processing.

Part C

Sleuth Kit Filesystem and Data-Unit Examination

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls -lh evidence/Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Sep  3 17:58 evidence/Ch01InChap01.dd

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ sha256sum evidence/Ch01InChap01.dd | tee hashes/Ch01InChap01.sha256
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122  evidence/Ch
01InChap01.dd

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ file evidence/Ch01InChap01.dd
evidence/Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ",'
k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes <=32 MB
), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by
descriptor+sectors), followed by FAT
```

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ img_stat evidence/Ch01InChap01.dd | tee working/Ch01InChap01_img_stat.txt
IMAGE FILE INFORMATION
--------------------------------------------
Image Type: raw

Size in bytes: 1474560
Sector size:    512

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ mmls evidence/Ch01InChap01.dd | tee working/Ch01InChap01_mmls.txt
```

Examine the FAT12 filesystem

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ fsstat evidence/Ch01InChap01.dd | tee working/Ch01InChap01_fsstat.txt
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT12

OEM Name: ,'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: �3��M|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
--------------------------------------------
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
--------------------------------------------
33-236 (204) -> EOF
285-311 (27) -> EOF
```

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ fls -r evidence/Ch01InChap01.dd | tee working/Ch01InChap01_fls.txt
r/r 5:	Client Info.mdb
r/r * 8:	Billing Letter.doc
r/r * 11:	confirmation.txt
r/r 13:	Income.xls
r/r * 15:	letter1.txt
r/r * 17:	Regrets.doc
v/v 45779:	$MBR
v/v 45780:	$FAT1
v/v 45781:	$FAT2
V/V 45782:	$OrphanFiles
```

Examine the deleted file metadata
From your fls output, the deleted files are:

8 — Billing Letter.doc
11 — confirmation.txt
15 — letter1.txt
17 — Regrets.doc

Let's examine inode 8 first.

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 8 | tee working/istat_8_Billing_Letter.txt
Directory Entry: 8
Not Allocated
File Attributes: File, Archive
Size: 24064
Name: _ILLIN~1.DOC

Directory Entry Times:
Written:        2005-12-09 06:50:28 (EST)
Accessed:       2005-12-09 00:00:00 (EST)
Created:        2005-12-09 06:59:05 (EST)

Sectors:
237 238 239 240 241 242 243 244
245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276
277 278 279 280 281 282 283

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 11 | tee working/istat_11_confirmation.txt
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR~1.TXT

Directory Entry Times:
Written:        2005-12-09 06:52:58 (EST)
Accessed:       2005-12-09 00:00:00 (EST)
Created:        2005-12-09 06:59:06 (EST)

Sectors:
284
```

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 15 | tee working/istat_15_letter1.txt
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:        2005-12-09 06:51:50 (EST)
Accessed:       2005-12-09 00:00:00 (EST)
Created:        2005-12-09 06:59:09 (EST)

Sectors:
312

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 17 | tee working/istat_17_Regrets.txt
Directory Entry: 17
Not Allocated
File Attributes: File, Archive
Size: 23552
Name: _EGRETS.DOC

Directory Entry Times:
Written:        2005-12-09 06:50:52 (EST)
Accessed:       2005-12-09 00:00:00 (EST)
Created:        2005-12-09 06:59:09 (EST)

Sectors:
313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328
329 330 331 332 333 334 335 336
337 338 339 340 341 342 343 344
345 346 347 348 349 350 351 352
353 354 355 356 357 358
```

Recover the deleted files with icat / Verify the recovered files

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 8 > recovered/Billing_Letter.doc

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 11 > recovered/confirmation.txt

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 15 > recovered/letter1.txt

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 17 > recovered/Regrets.doc

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls -lh recovered/
total 56K
-rw-rw-r-- 1 kali kali 24K Sep  3 18:23 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 227 Sep  3 18:24 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Sep  3 18:24 letter1.txt
-rw-rw-r-- 1 kali kali 23K Sep  3 18:24 Regrets.doc

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ file recovered/*
recovered/Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title:
 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creatin
g Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri
Dec  9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
recovered/confirmation.txt:   ASCII text, with CRLF line terminators
recovered/letter1.txt:        ASCII text, with CRLF line terminators
recovered/Regrets.doc:        Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title:
 02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creati
ng Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri
 Dec  9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0
```

Examine the recovered text

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ cat recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroiu9

Thank you for your business

George

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ cat recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
```

Block-level examination

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ blkcat evidence/Ch01InChap01.dd 284 | tee working/blkcat_sector_284.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroiu9

Thank you for your business

George

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ blkcat evidence/Ch01InChap01.dd 312 | tee working/blkcat_sector_312.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ blkcat evidence/Ch01InChap01.dd 237 | tee working/blkcat_sector_237.txt
��▯▮�>��        *,����)������������������������������������������������������������������������������
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┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ▮
```

The forensic image Ch01InChap01.dd was examined using The Sleuth Kit. The image was identified as a raw FAT12 filesystem with a size of 1,474,560 bytes and a 512-byte sector size. The **SHA-256 hash was recorded as 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122**.

Filesystem analysis identified four deleted/unallocated files: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. istat confirmed that their directory entries were marked Not Allocated while their associated sectors remained available.

The files were recovered successfully using icat. The recovered Word documents were identified as valid Microsoft Word Compound Document files, while the text files were identified as ASCII text. blkcat was also used to examine individual data sectors, confirming that sector-level content corresponded to the recovered deleted files.

The recovered text contained communications involving Laura, George, and Earl, including an FTP-related message and a separate meeting-related message. These findings demonstrate that deleted file contents can remain recoverable even after filesystem directory entries have been marked unallocated.

Part D

Binwalk

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ cd ~/SBT-DF202_Lab3

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ binwalk J_ub_law.jpg | tee working/J_ub_law_binwalk.txt

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             JPEG image data, EXIF standard
12            0xC             TIFF image data, little-endian offset of first image directory: 8
26844         0x68DC          Copyright string: "Copyright 1999 Adobe Systems Incorporated"

┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ █
```

Embedded Content Analysis Using Binwalk

Binwalk was used to scan J_ub_law.jpg for embedded file signatures and additional data structures. The scan identified JPEG/EXIF data at offset 0, TIFF data at offset 12 within the EXIF structure, and an Adobe copyright string at offset 26,844. No separate embedded archive or additional recoverable file was identified.

Finding: The Binwalk results are consistent with a normal JPEG image containing EXIF/TIFF metadata and Adobe-related metadata. No additional embedded file requiring extraction was detected.

Part E Prepare and Examine the USB Image (120M.7z).

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ ls -lh 120M.7z
-rw-rw-r-- 1 kali kali 36M Sep  3 18:59 120M.7z
```

Compute Hashes

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ hashdeep -c md5,sha1 120M.7z
%%%% HASHDEEP-1.0
%%%% size,md5,sha1,filename
## Invoked from: /home/kali/SBT-DF202_Lab3/evidence
## $ hashdeep -c md5,sha1 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2810745018afaa2da31dc17a8eb590fca66eeef7,/home/kali/SBT-DF202_Lab3/evidence/120M.7z

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ █
```

List the content of the zipfile

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ 7z l 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
 64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

   Date      Time    Attr         Size   Compressed  Name
------------------- ----- ------------ ------------  ------------------------
2021-09-22 22:59:31 D....A            0            0  120M
2021-09-22 22:59:31 ....A    124780544     36720256  120M/usb_fat_carving.001
2021-09-22 22:59:32 ....A         1685               120M/usb_fat_carving.001.txt
------------------- ----- ------------ ------------  ------------------------
2021-09-22 22:59:32          124782229     36720256  2 files, 1 folders
```

inspect the accompanying .txt

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ 7z x 120M.7z "120M/usb_fat_carving.001.txt" -so | tee ../working/usb_fat_carving_notes.txt
Created By AccessData® FTK® Imager 4.5.0.3

Case Information:
Acquired using: ADI4.5.0.3
Case Number: USB File Carving
Evidence Number: UB-002
Unique description: USB files used for file carving
Examiner: Frank Xu
Notes: University of Baltimore

--------------------------------------------------------------

Information for D:\cloud_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ_Team_Repository\DOJ_Case_Study\0_Basic_Com
puter_Skills_for_Forensics\file_carving\usb_image\120M\usb_fat_carving:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 15
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 243,712
[Physical Drive Information]
 Drive Model: General UDisk USB Device
 Drive Serial Number:
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 119 MB
 Sector count:    243712
[Computed Hashes]
 MD5 checksum:    ba4a1d0ba49f4a6667b00a3b3e85e604
 SHA1 checksum:   bcc2d49fd49c9521ecb1739f6542c6bf327375ef

Image Information:
 Acquisition started:   Wed Sep 22 22:59:16 2021
 Acquisition finished:  Wed Sep 22 22:59:31 2021
 Segment list:
  D:\cloud_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ_Team_Repository\DOJ_Case_Study\0_Basic_Computer_Skills_f
or_Forensics\file_carving\usb_image\120M\usb_fat_carving.001

Image Verification Results:
 Verification started:  Wed Sep 22 22:59:32 2021
 Verification finished: Wed Sep 22 22:59:32 2021
 MD5 checksum:    ba4a1d0ba49f4a6667b00a3b3e85e604 : verified
 SHA1 checksum:   bcc2d49fd49c9521ecb1739f6542c6bf327375ef : verified
```

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ 7z e 120M.7z "120M/usb_fat_carving.001" -o.

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
 64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Size:        124780544
Compressed: 36720470

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ ls -lh usb_fat_carving.001
-rw-rw-r-- 1 kali kali 119M Sep 22  2021 usb_fat_carving.001
```

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ sha256sum usb_fat_carving.001 | tee ../hashes/usb_fat_carving.001.sha256
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9  usb_fat_carving.001

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ file usb_fat_carving.001
usb_fat_carving.001: DOS/MBR boot sector MS-MBR Windows 7 english at offset 0×163 "Invalid partition table" at offset 0×17b "Error l
oading operating system" at offset 0×19a "Missing operating system", disk signature 0×a1159a00; partition 1 : ID=0×e, active, start-
CHS (0×0,2,3), end-CHS (0×e,254,63), startsector 128, 243584 sectors

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ img_stat usb_fat_carving.001
IMAGE FILE INFORMATION
--------------------------------------------
Image Type: raw

Size in bytes: 124780544
Sector size:    512

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ mmls usb_fat_carving.001
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000000127   0000000128   Unallocated
002:  000:000   0000000128   0000243711   0000243584   Win95 FAT16 (0×0e)
```

What we confirmed
SHA-256: 9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9
Image type: Raw forensic image
Sector size: 512 bytes
Total size: 124,780,544 bytes
Partition table: DOS/MBR
Partition: Win95 FAT16 (0x0e)
Partition start: Sector 128
Partition length: 243,584 sectors
Sectors 0–127 are unallocated before the partition.

The important point is that the FAT16 filesystem does not start at sector 0. It starts at sector 128, so we need to account for that offset when examining the filesystem.

Examine the the content of the usb

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ fsstat -o 128 usb_fat_carving.001 | tee ../working/usb_fat_fsstat.txt
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT16

OEM Name: MSDOS5.0
Volume ID: 0x94105672
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): USB
File System Type Label: FAT16

Sectors before file system: 128

File System Layout (in sectors)
Total Range: 0 - 243583
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 241
* FAT 1: 242 - 479
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583

METADATA INFORMATION
--------------------------------------------
Range: 2 - 3889670
Root Directory: 2

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 60769

FAT CONTENTS (in sectors)
--------------------------------------------
512-515 (4) -> EOF
516-519 (4) -> EOF
520-523 (4) -> EOF
70976-71211 (236) -> EOF
71212-71215 (4) -> EOF
```

Display Partitions

```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                Boot Start    End Sectors   Size Id Type
usb_fat_carving.001p1 *      128 243711  243584 118.9M  e W95 FAT16 (LBA)
```

Find the delete files

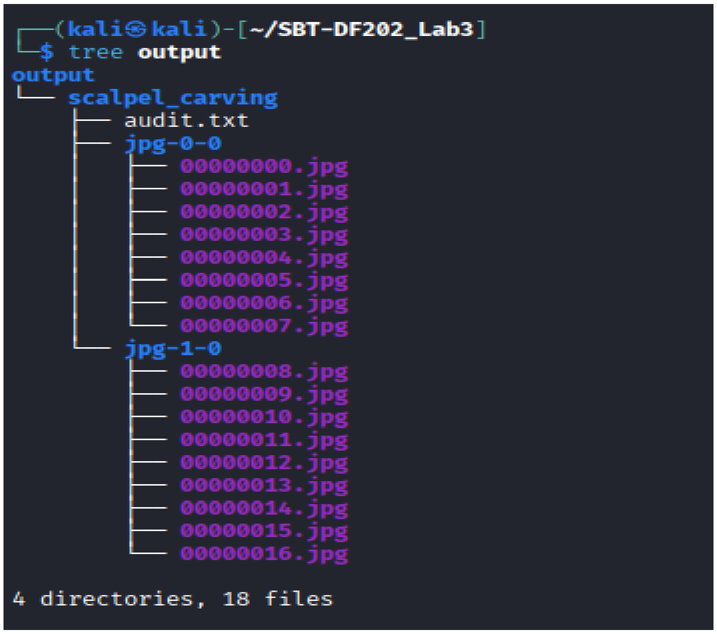
```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ fls -o 128 -r usb_fat_carving.001 | tee ../working/usb_fat_fls.txt
r/r 3:  USB         (Volume Label Entry)
d/d 6:  System Volume Information
+ r/r 519:      WPSettings.dat
r/r * 7:        _est
r/r 10: .dropbox.device
d/d * 13:       old_File_Carving_files
+ r/r * 711:    CarvedImage.jpg
+ r/r * 714:    File_carving.docx
+ r/r * 717:    File_Carving.pptx
+ r/r * 720:    i-love-cats-cute-cats.jpg
r/r * 15:       B_ub_poe4.bmp
r/r * 18:       B_zoom-eubie-mono.bmp
r/r * 21:       Ballardlab8.java
r/r * 24:       brittLab10.java
r/r * 27:       DO_example.doc
r/r * 30:       DO_example2.doc
r/r * 33:       G_BuiltForThis.gif
r/r * 35:       G_zoom-sc.gif
r/r * 37:       H_Form.html
r/r * 39:       H_hello.html
r/r * 41:       J_ub_law.jpg
r/r * 44:       J_ub_night.jpg
r/r * 47:       nps-2008-jean_outlook.pst
r/r * 50:       P_CAS-zoom-6.png
r/r * 53:       P_MSB_1_zoom.png
r/r * 57:       pd_Evidence_search_techniques.pdf
r/r * 61:       pd_Forensic_Report_Template.pdf
r/r * 64:       pp_Number_Systems.pptx
r/r * 67:       pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71:       _tf_1.rtf
r/r * 74:       T_Eubie-iphone-5-8.tiff
r/r * 78:       T_youknowus-iphone-5-8.tiff
r/r * 81:       W_axloadcomplete.wav
r/r * 84:       W_speaker_test_sound.wav
r/r * 86:       Z_file.zip
r/r * 88:       江莎莎行踪轨迹.doc
r/r * 91:       nps-2008-jean_outlook.pst
v/v 3889667:    $MBR
v/v 3889668:    $FAT1
v/v 3889669:    $FAT2
V/V 3889670:    $OrphanFiles
```

Part F — Scalpel File Carving

Carving usb image

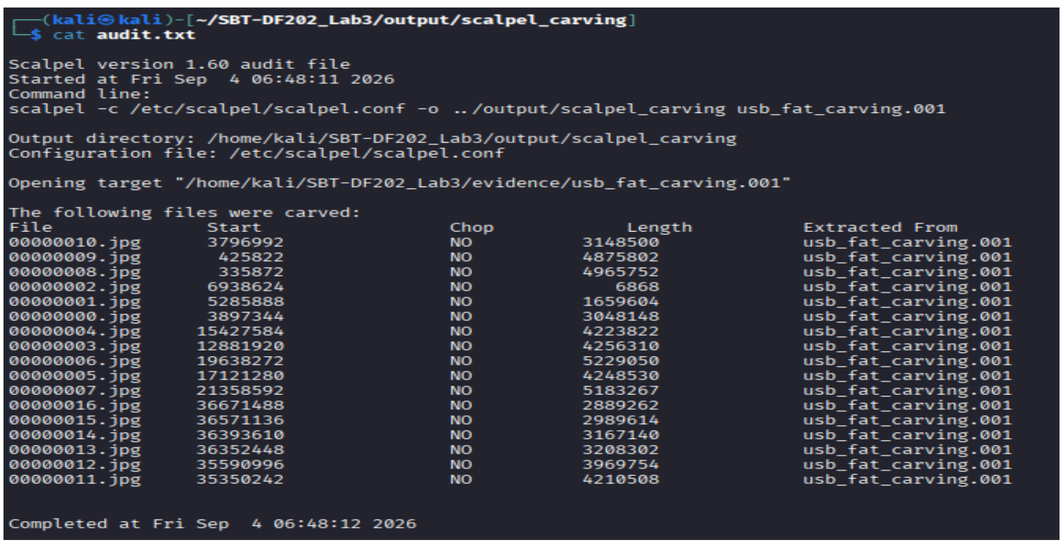
```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ sudo mousepad /etc/scalpel/scalpel.conf

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ scalpel -c /etc/scalpel/scalpel.conf -o ../output/scalpel_carving usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.
ERROR: You have attempted to use a non-empty output directory. In order
       to maintain forensic soundness, this is not allowed.
Aborting.


┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ rm -rf ../output/scalpel_carving
mkdir -p ../output/scalpel_carving

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ scalpel -c /etc/scalpel/scalpel.conf -o ../output/scalpel_carving usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/SBT-DF202_Lab3/evidence/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*******************************************************************************|  119.0 MB    00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built.  Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" --> 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" --> 9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*******************************************************************************|  119.0 MB    00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 1 seconds.

┌──(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
└─$ 
```

Show carved files



```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ tree output
output
└── scalpel_carving
    ├── audit.txt
    ├── jpg-0-0
    │   ├── 00000000.jpg
    │   ├── 00000001.jpg
    │   ├── 00000002.jpg
    │   ├── 00000003.jpg
    │   ├── 00000004.jpg
    │   ├── 00000005.jpg
    │   ├── 00000006.jpg
    │   └── 00000007.jpg
    └── jpg-1-0
        ├── 00000008.jpg
        ├── 00000009.jpg
        ├── 00000010.jpg
        ├── 00000011.jpg
        ├── 00000012.jpg
        ├── 00000013.jpg
        ├── 00000014.jpg
        ├── 00000015.jpg
        └── 00000016.jpg

4 directories, 18 files
```

Show audit log



```
┌──(kali㉿kali)-[~/SBT-DF202_Lab3/output/scalpel_carving]
└─$ cat audit.txt

Scalpel version 1.60 audit file
Started at Fri Sep  4 06:48:11 2026
Command line:
scalpel -c /etc/scalpel/scalpel.conf -o ../output/scalpel_carving usb_fat_carving.001

Output directory: /home/kali/SBT-DF202_Lab3/output/scalpel_carving
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/kali/SBT-DF202_Lab3/evidence/usb_fat_carving.001"

The following files were carved:
File              Start                 Chop         Length          Extracted From
00000010.jpg      3796992               NO          3148500          usb_fat_carving.001
00000009.jpg       425822               NO          4875802          usb_fat_carving.001
00000008.jpg       335872               NO          4965752          usb_fat_carving.001
00000002.jpg      6938624               NO             6868          usb_fat_carving.001
00000001.jpg      5285888               NO          1659604          usb_fat_carving.001
00000000.jpg      3897344               NO          3048148          usb_fat_carving.001
00000004.jpg     15427584               NO          4223822          usb_fat_carving.001
00000003.jpg     12881920               NO          4256310          usb_fat_carving.001
00000006.jpg     19638272               NO          5229050          usb_fat_carving.001
00000005.jpg     17121280               NO          4248530          usb_fat_carving.001
00000007.jpg     21358592               NO          5183267          usb_fat_carving.001
00000016.jpg     36671488               NO          2889262          usb_fat_carving.001
00000015.jpg     36571136               NO          2989614          usb_fat_carving.001
00000014.jpg     36393610               NO          3167140          usb_fat_carving.001
00000013.jpg     36352448               NO          3208302          usb_fat_carving.001
00000012.jpg     35590996               NO          3969754          usb_fat_carving.001
00000011.jpg     35350242               NO          4210508          usb_fat_carving.001


Completed at Fri Sep  4 06:48:12 2026
```

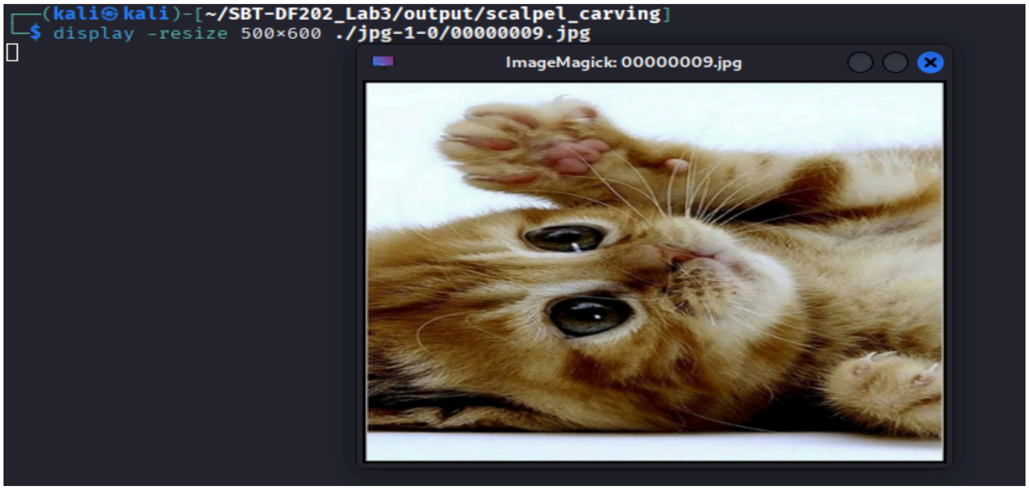
Scalpel Carving Results

The Scalpel audit log documented 17 JPEG files carved from usb_fat_carving.001. The carved files were assigned sequential names from 00000000.jpg through 00000016.jpg.

The recorded starting offsets ranged from 335,872 bytes to 36,671,488 bytes, with extracted lengths ranging from 6,868 bytes to 5,229,050 bytes. All 17 files were reported with Chop: NO.

The Scalpel audit file provides a repeatable record of the carving operation, including the source image, configuration file, output directory, file offsets, and extracted lengths.

Finding: JPEG data was successfully recovered directly from the raw USB image using file signatures, demonstrating recovery independent of normal filesystem metadata.



Scalpel successfully carved 17 JPEG files from the USB forensic image. The recovered images were stored in the jpg-0-0 and jpg-1-0 directories. Two carved JPEG files were opened for visual examination, demonstrating successful recovery of usable image content from the raw forensic image.

Analysis and Findings

The forensic analysis produced the following findings:

1. JPEG Hexadecimal Analysis

The JPEG sample J_ub_law.jpg was examined using xxd. The JPEG Start of Image (SOI) signature FF D8 and End of Image (EOI) signature FF D9 were identified. The hexadecimal data was successfully reconstructed into a JPEG file. The reconstructed file produced identical MD5 and SHA-256 hashes to the original, confirming bit-for-bit integrity.

2. JPEG Metadata Analysis

ExifTool identified the image as a JPEG captured using a NIKON D4 camera. Embedded metadata included timestamps, camera information, copyright information, keywords, and Adobe Photoshop processing history. These metadata elements provided useful investigative information but were treated cautiously because metadata can be altered.

3. Deleted-File Recovery

The FAT12 forensic image Ch01InChap01.dd was analyzed using The Sleuth Kit. Four deleted files were identified: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The files were marked as Not Allocated but their content remained recoverable using icat.

4. Sector-Level Analysis

blkcat was used to examine individual sectors associated with deleted files. Readable text was recovered from sectors containing confirmation.txt and letter1.txt, demonstrating that file content can remain present even after filesystem entries have been deleted.

5. Binwalk Analysis

Binwalk identified JPEG/EXIF data, TIFF data within the EXIF structure, and an Adobe copyright string. No separate embedded archive or additional recoverable file was identified.

6. USB Image and Scalpel Analysis

The USB forensic image was identified as a FAT16 filesystem. The Sleuth Kit identified multiple deleted files and directories. Scalpel was configured to recognize JPEG files using Exif and JFIF signatures.

Scalpel successfully carved 17 JPEG files from the raw USB image. The recovery was documented in audit.txt, including the starting offsets and extracted lengths of the carved files. Two recovered JPEG files were subsequently opened for visual examination.

Overall Finding

The investigation demonstrated that deleted or fragmented digital evidence may remain recoverable from raw storage media even when normal filesystem metadata is unavailable or marked as unallocated. The combination of filesystem analysis, hexadecimal examination, metadata analysis, sector-level recovery, and signature-based carving provided multiple methods for identifying and recovering forensic evidence.

Conclusion

This laboratory exercise successfully demonstrated the application of data carving and file recovery techniques in digital forensics. Hexadecimal analysis, metadata examination, filesystem analysis, sector-level examination, embedded-content scanning, and signature-based file carving were used to identify and recover digital evidence.

The investigation successfully reconstructed and verified a JPEG file, recovered deleted files from a FAT12 forensic image, analyzed a FAT16 USB image, and used Scalpel to carve 17 JPEG files from raw storage data. Cryptographic hashing and audit logs were also used to support evidence integrity and reproducibility.

Overall, the exercise demonstrated that deleted files and useful digital evidence can remain recoverable from storage media even when filesystem metadata is missing or marked as unallocated. The results highlight the importance of combining multiple forensic techniques while preserving the integrity of the original evidence.